

Technical Documentation

Scrapy Project

The [Scrapy](#) project is designed to retrieve and store 14-day weather forecast data for multiple locations globally. The system is implemented using [Scrapy](#) and [Selenium](#) for data extraction, a PostgreSQL database for storage, and Docker for containerization and deployment on DigitalOcean. [ScrapeOps](#) provides a dashboard and rotating proxies and browser agents.

Architecture

1. Web Scraping Layer

- Spiders (Scrapy):
 - [MeteoProg](#) Spider: Scrapes 14-day weather forecast data, focusing on day-to-day continuity and avoiding specific hour-based comparisons.
 - [MeteoBlue](#) Spider: Utilizes Selenium for dynamic data extraction from JavaScript-heavy websites.
 - [TheWeatherChannel](#) Spider: Extracts weather details, including temperatures and precipitation chances, using precise XPath selectors.
 - [TimeAndDate](#) Spider: Utilizes Selenium for dynamic data extraction from JavaScript-heavy websites.

2. Middleware Layer

ScrapeOps Middleware:

- Integrates with the **Scrapy** framework to manage and rotate proxies, set custom browser headers, and manage user agents.
- **Rotating Proxies**: ScrapeOps provides a pool of rotating proxies to prevent blocking or throttling by websites.
- **User Agents & Headers**: Ensures the scraping requests appear to come from a diverse set of real browsers by rotating headers and user agents for each request, improving the spider's success rate.

3. Data Processing Layer

- Pipelines: Saves data to PostgreSQL.

4. Database Layer

- PostgreSQL: Stores normalized weather data for efficient querying and analysis.

- Integration with spiders via Scrapy pipelines.
- Database is secured by IP whitelisting restriction to VPN server, Scrapy server and API.